**Task Sensitivity Modulates Implicit Causality Effects Revealed by Noun Gender Bias**

Implicit causality (IC) concerns a well-studied class of verbs that can guide pronoun interpretation, by implicating one participant as the driving force behind an event's cause, thus biasing this participant to be re-mentioned in an explanation (e.g., *Tom annoyed Mark because he…he=Tom*)**[1,2]**. Recent sentence-completion work suggests that English noun gender biases can outweigh IC biases, with comprehenders more often selecting the gender-matching noun as a pronoun's referent even when it conflicts with IC bias**[3,4]**. However, how IC bias interacts with noun gender during online sentence processing remains unclear. By integrating work on how noun gender bias impacts pronoun resolution**[5]** with research on the time course of IC effects in sentence processing**[6-8]**, this project assesses how these two sources of information jointly shape online anaphor resolution. The integration account holds that IC influences only later stages of interpretation, when comprehenders integrate information across clauses**[6]**. Under this view, in ***Linda*** *annoyed David because* ***he*** *was late*, the gender-mismatching pronoun (*he*) should not trigger early processing difficulty due to IC conflict. Effects should emerge only after clause-level integration. In contrast, the immediate focusing account proposes that IC information guides attention as soon as the verb is processed, predicting early difficulty when a pronoun's gender mismatches the IC-favored noun**[7,8]**. Prior findings are mixed. Some self-paced reading (SPR) studies report no early IC effects**[6]**, while others using SPR, eye-tracking, or the Visual World Paradigm report earlier IC activation**[7,8]**. But even these studies show effects mainly in spillover regions or later eye-tracking measures, leaving the timing of IC bias unresolved. The present work addresses questions both in the domain of noun gender bias and the time course of IC information by comparing two experimental tasks that differ in effect localization: SPR, which is prone to spillover**[i.a.,9]**, and the Grammatical Maze (G-Maze) task, which is argued to yield more localized difficulty measures**[10-13]**.

**Exp. 1: SPR.** Participants (N=130) saw items word-by-word using SPR. Stimuli had a 2x2 design: (semantic) Noun Gender (fem vs. masc, IC biased NP gender bias) and Pro Match (match vs. mismatch, reflecting whether the pronoun matches in gender with the IC biased NP), see (1). All IC verbs were subject biasing, selected from previous work showing robust IC effects**[14-16]**. A linear mixed effects regression model was fit, predicting RTs at the pronoun, and a separate model predicting RTs at the pronoun spillover region. We did not find a main effect of Pronoun Match at the pronoun ($p=.39$) or spillover ($p=.29$) regions – see Fig. 1&2.

**Exp. 2: G-Maze.** Participants (N=80) saw items word-by-word using the G-Maze task in which they needed to distinguish the correct continuation from a distractor. Stimuli and statistical analysis was the same as Exp. 1. For this experiment, we found a main effect of Pronoun Match at the pronoun region ($p < .001$***) – see Fig. 3&4. We also ran SPR/G-Maze with stereotypically gendered nouns (e.g., *beautician*, *plumber*) but did not find any meaningful differences between noun types. **Taken together,** these findings speak to both gender-bias research and debates about the timing of IC effects**.** Gender-biased nouns generate strong pronoun expectations, allowing us to detect conflicts with IC-driven predictions during online processing. The G-Maze slowdown for pronoun mismatch shows that participants anticipated the IC-biased noun as the referent, revealing an early IC influence consistent with the immediate focusing account. The contrast between the null SPR results and the robust G-Maze effect further indicates that detecting such early IC effects depends on the sensitivity and localization of the processing measure, with G-Maze revealing biases that are obscured in SPR.

**References:**

[1] Caramazza et al., 1977 [2] Kehler et al., 2008 [3] Kaiser & Soto, 2025 [4] Davenport & Ronai 2025 [5] Sturt 2003 [6] Stewart et al. 2000 [7] Koorneef & Van Berkum, 2005 [8] Pyykkönen & Järvikivi 2009 [9] Roland et al., 2021 [10] Boyce et al., 2020 [11] Forster et al. 2009 [12] Witzel et al. 2012 [13] Zehr and Schwarz 2022 [14] Au, 1986 [15] Hartshorne & Snedeker, 2013 [16] McKoon et al., 1993

**Figures:**

**Fig. 1. SPR: RTs on Pro Region**

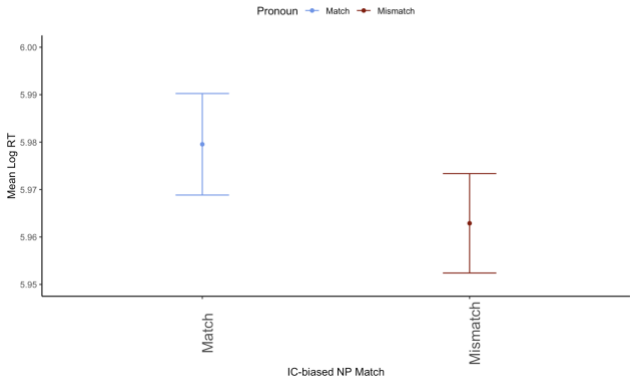



**Fig. 3. G-Maze: RTs on Pro Region**

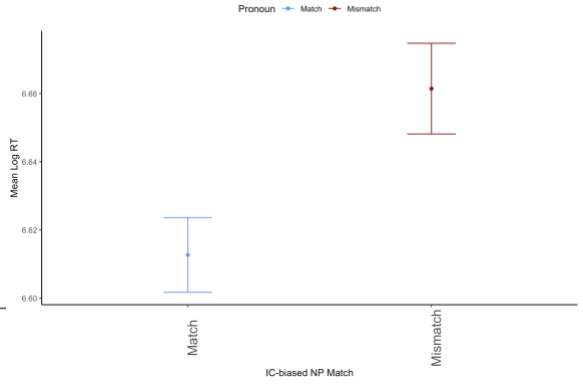



**Fig. 2. SPR: RTs on Spill Region**

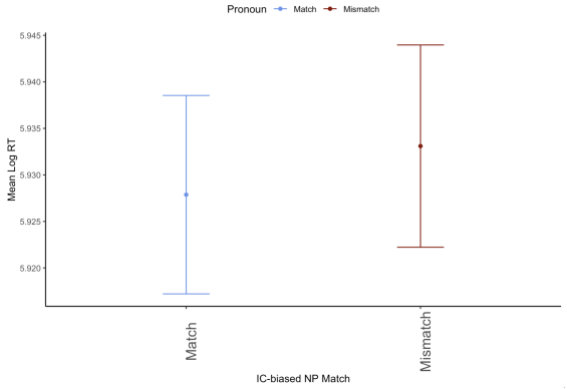
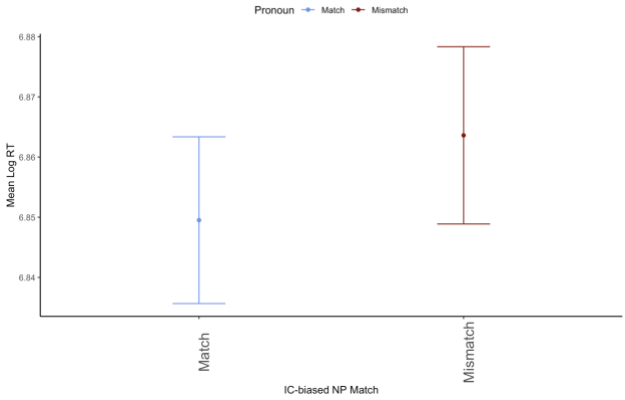



**Fig. 4. G-Maze: RTs on Spillover Region**




**(1) Stimuli Example:**

a. The / **bridesmaid** / annoyed / the / groom / because / **she** / truly / had / no / patience. [Fem, Pro Match]
b. The / **bridesmaid** / annoyed / the / groom / because / **he** / truly / had / no / patience. [Fem, Pro Mismatch]
c. The / **groom** / annoyed / the / bridesmaid / because / **he** / truly / had / no / patience. [Male, Pro Match]
d. The / **groom** / annoyed / the / bridesmaid / because / **she** / truly / had / no / patience. [Male, Pro Mismatch]